



Anti-apoptotic effect of modified Chunsimyeolda-tang, a traditional Korean herbal formula, on MPTP-induced neuronal cell death in a Parkinson's disease mouse model



Huan Li^{a,d,e,1}, Gunhyuk Park^{b,1}, Nayoung Bae^{a,c}, Joonki Kim^a, Myung Sook Oh^{b,*}, Hyun Ok Yang^{a,d,**}

^a Natural Products Research Center, Korea Institute of Science and Technology, Gangneung 210-340, Republic of Korea

^b Department of Life and Nanopharmaceutical Science and Kyung Hee East-West Pharmaceutical Research Institute, Kyung Hee University, Seoul 130-701, Republic of Korea

^c Department of Sasang Constitution Medicine, Pusan National University School of Korean Medicine, Yangsan 626-870, Republic of Korea

^d Department of Biological Chemistry, University of Science & Technology (UST), Daejeon 305-350, Republic of Korea

^e College of Chemistry and Chemical Engineering, Qiqihar University, Qiqihar 161006, China

ARTICLE INFO

Article history:

Received 22 June 2015

Received in revised form

1 November 2015

Accepted 4 November 2015

Available online 23 November 2015

Keywords:

Modified-Chungsimeolda-tang

Parkinson's disease

Neuroprotection

Apoptosis

1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine

6-Hydroxydopamine

ABSTRACT

Ethnopharmacological relevance: The modified-Chungsimeolda-tang (DG) is an important traditional Korean herbal formula used in traditional oriental medicine for treatment of cerebrovascular disorders, including stroke. The formula is based on the book “Dongui Sasang Shinpyun”.

Aim of the study: In the previous studies, the neuroprotective effect of DG is demonstrated in an *in vitro* Parkinson's disease (PD) model, and in this study, the 1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) animal model of PD is used to evaluate the behavioral effect of DG and possible mechanism through anti-apoptosis of DG. 6-Hydroxydopamine (6-OHDA) also is used to evaluate the anti-apoptosis effect of DG in SH-SY5Y cells.

Materials and methods: MPTP was used to evaluate the behavioral damage and neurotoxicity in mice. The bradykinesia symptom was measured by a Pole test and a Rota-rod test in mice. Also the loss of tyrosine hydroxylase (TH)-positive neurons induced by MPTP was examined by an immunohistochemical assay. The DG-mediated anti-apoptosis effect was measured using an immunoblotting assay with apoptosis-related markers such as Bax and cleaved caspase-3. DG and 1-methyl-4-phenylpyridinium (MPP⁺) were co-treated with primary dopaminergic neurons to evaluate the protective effect of DG. The expression of caspase-3 and PARP was measured to detect the protective effect of DG from the damage by 6-OHDA.

Results and conclusions: The treatment with DG resulted in prophylactic effects on MPTP-induced Parkinsonian bradykinesia and the immunohistochemical analysis showed that DG provided the neuroprotection against the MPP⁺-induced dopaminergic neurons loss through the anti-apoptosis effect. The present results suggested that it might be possible to use DG for the prevention of substantia nigra pars compacta (SNpc) degeneration induced by exposure to the toxic substances, such as MPTP/MPP⁺, in PD mouse model.

© 2015 Elsevier Ireland Ltd. All rights reserved.

Abbreviations: ABC, avidin-biotin peroxidase complex; DG, modified Chungsimeolda-tang; FBS, fetal bovine serum; MYH, modified Youldahanso-tang; PD, Parkinson's disease; MEM, minimal essential medium; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; MPP⁺, 1-methyl-4-phenylpyridinium; T-LA, time for locomotion activity; ROS, reactive oxygen species; ST, striatum; SNpc, substantia nigra pars compacta; TH, tyrosine hydroxylase; 6-OHDA, 6-hydroxydopamine

* Corresponding author.

** Corresponding author at: Natural Products Research Center, Korea Institute of Science and Technology, Gangneung 210-340, Gangwon-do, Republic of Korea.

E-mail addresses: msohok@khu.ac.kr (M.S. Oh), hoyang@kist.re.kr (H.O. Yang).

¹ These two authors contributed equally to this work.

1. Introduction

Parkinson's disease (PD), the second most common neurodegenerative disorder after Alzheimer's disease, is characterized by motor and behavioral disturbances that include a resting tremor, postural instability, and bradykinesia (Shimohama et al., 2003). The pathological hallmark of PD is degeneration of Dopaminergic neurons in the substantia nigra pars compacta (SNpc) and a subsequent loss of dopamine in the striatum (ST) (Bove et al., 2005; Shimohama et al., 2003).

It is widely agreed that oxidative stress plays a pivotal role in the neurodegeneration associated with PD (Chakraborty et al.,

2013; Subramaniam and Chesselet, 2013; Zuo and Motherwell, 2013). Oxidative stress is known to damage lipids, proteins, and DNA and these, along with decreased glutathione levels, have been observed in the postmortem ST and SNpc of PD (Schapira and Jenner, 2011). Data from human postmortem tissue also indicate that alterations in reactive oxygen species (ROS), nitric oxide, and mitochondrial complex I activity are important in the pathogenesis of sporadic PD (Banerjee et al., 2009).

As a highly lipophilic molecule, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) readily crosses the blood–brain barrier within minutes after systemic administration (Markey et al., 1984), and then converts into 1-methyl-4-phenylpyridinium (MPP⁺), which is believed to elicit a specific intoxication of dopaminergic neurons (Dauer and Przedborski, 2003). MPTP can selectively damage dopaminergic neurons, which has been used extensively in various animal species. MPTP can induce behavioral, neuropathological and biochemical changes in primates and mice, which are similar to those that occur in idiopathic PD (Bergman and Deuschl, 2002; Przedborski et al., 2000). The MPTP mouse model is a common method used to study the neuroprotective effect of drugs and has provided considerable therapeutic insight into basal ganglia physiology and response to drug therapy (Smeyne and Jackson-Lewis, 2005). This model can recapitulate the primary pathological and biochemical features of Parkinson's disease, such as oxidative stress, mitochondrial dysfunction, and apoptosis (Schmidt and Ferger, 2001).

Cellular apoptotic dysfunction is one of the triggering events that lead to neuronal cell death with PD. It is now believed that apoptosis results from activated caspase proteolysis of various cellular components initiated by ROS generation (Serviddio et al., 2011). The mechanism of apoptosis is complex and involves a cascade of reactions; one of the key steps leading to apoptosis is the leakage of cytochrome C from the mitochondria and activation of caspase-3 (Wang et al., 2014). Hartmann et al. (2000) reported that using an antibody raised against activated caspase-3, the percentage of active caspase-3-positive neurons among dopaminergic neurons was significantly higher in PD patients than in controls (Hartmann et al., 2000). In another human PD brain study, it is observed that a significantly higher percentage of dopaminergic SN neurons displayed caspase-8 activation in PD patients compared with controls (Hartmann et al., 2001). Two other studies found an increased expression of the anti-apoptotic factor Bcl-2 in the SNpc of PD patients compared to the cerebral cortex and age matched controls, probably as a compensatory mechanism (Marshall et al., 1997; Mogi et al., 1996). Therefore, suppressing the key elements related to the induction of apoptosis may propose a possible therapeutic approach in PD.

The modified Chungsimyeolda-tang (DG), a traditional Korean herbal formula, has been described in 'Dongui Sasang Shinpyun', a book that has a great effect on the clinical utilization of 'Sasang Constitutional Medicine' prescriptions. Sasang Constitutional Medicine, first introduced by Jema Lee in 1894, is an indispensable part of traditional Korean medicine (Leem and Park, 2007; Shim et al., 2008). The major ingredients of DG are *Polygala tenuifolia* (Willd.), *Angelica tenuissima* Nakai and *Dimocarpus longan* Lour mixed at 1:1:1 ratio (dry weight). The *Polygala tenuifolia* (Willd.) was used extensively in neurodegeneration diseases (Choi et al., 2011; Lv et al., 2009; More et al., 2013; Naito and Tohda, 2006). The *Angelica tenuissima* Nakai and *Dimocarpus longan* Lour were considered to have anti-oxidative, anti-cancer and anti-microbial activities (Jiang et al., 2014; Ka et al., 2005; Tseng et al., 2014) related with various diseases, including neurodegenerative disease (Barrett and Timothy Greenamyre, in press; Jazvinscak Jembrek et al., 2015; Romanucci and Della Salda, 2015). In our previous studies, we have used an *in vitro* PD model, exposure to toxic substances-MPP⁺ and rotenone, to evaluate the neuro-protective

effect of DG in SH-SY5Y cells (Bae et al., 2015), and also we have confirmed the protective effect of DG through autophagy induction in *in vitro*. The increasing evidence indicated that the autophagy has the emerging role in the Parkinson's disease, the pathogenesis of which appeared to be converging on oxidative stress, mitochondrial dysfunction and protein aggregation, all of which are tightly linked to autophagy (Lynch-Day et al., 2012). And the aim of the present study was to investigate the therapeutic potential of DG in experimental *in vivo* PD models. In this study, we evaluated the neuro-protective effects of DG against MPTP-induced neurotoxicity and the possible mechanisms of its effects *in vivo* were examined by performing behavioral impairment tests, brain tissue stereology, and measurements of apoptotic proteins in an experimental animal PD model. In addition, the neuro-protective effect of DG and its fractions was evaluated by exposure to 1-methyl-4-phenylpyridinium (MPP⁺) to induce toxicity in primary dopaminergic neurons.

2. Materials and methods

2.1. Chemicals and reagents

Minimal essential medium (MEM), fetal bovine serum (FBS), horse serum and penicillin–streptomycin were purchased from Gibco (Carlsbad, CA, USA). Poly-L-lysine, MPTP, MPP⁺, 3,3-diaminobenzidine (DAB), and Bovine serum albumin (BSA) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Dulbecco's modified Eagle's medium (DMEM) and fetal bovine serum (FBS) were obtained from Hyclone (Logan, UT, USA). Rabbit anti-tyrosine hydroxylase (TH) antibody was purchased from Chemicon International Inc. (Temecula, CA, USA). Rabbit anti-cleaved caspase-3 antibody was obtained from Chemicon International Inc. (Temecula, CA, USA). Rabbit anti-Bax and mouse anti-β-actin antibodies were obtained from Santa Cruz Biotechnology Inc. (Santa Cruz, CA, USA). Rabbit anti-caspase-3, anti-PARP and anti-GAPDH antibodies were purchased from Cell Signaling Technology Inc. (Boston, MA, USA). Normal goat serum and an avidin-biotin peroxidase complex (ABC) standard kit were purchased from Vector Laboratories Inc. (Burlingame, CA, USA). Anti-rabbit and anti-mouse horseradish peroxidase secondary antibodies were purchased from Assay Designs Inc. (Ann Arbor, MI, USA). All of the other chemicals were of analytical grade.

2.2. Preparation of DG

DG consisted of three individual herbs: *Polygala tenuifolia* (Willd.), *Angelica tenuissima* Nakai and *Dimocarpus longan* Lour. DG was prepared according to our previous study (Bae et al., 2015). The herbs were purchased by the Oriental Hospital at Daejeon University (Daejeon, Republic of Korea). The botany and drug department of Oriental Hospital assumed the identification and authentication of the herbs. The voucher specimens of the herbs (voucher number: *Polygala tenuifolia* (Willd.), KN00847A; *Angelica tenuissima* Nakai, KN00735A and *Dimocarpus longan* Lour, KN00804A) were deposited at the herbarium, Korea Institute of Science & Technology (Gangneung, Republic of Korea). Briefly, *Polygala tenuifolia* (Willd.), *Angelica tenuissima* Nakai and *Dimocarpus longan* Lour were air-dried, weighed 15 g (dry weight) of each herbs and mixed together, total 45 g of crude drugs. Distilled water was added at final 1 L, and boiled for twice. A final volume of 100 mL solution was left after evaporated, labeled as the DG extract, then, centrifuged the extract and collected the supernatant. The supernatant was filtered, freeze dried and obtained 2.925 g lyophilized extract (the yield was 6.5%, which was from 2.925 g lyophilized extract/45 g crude drug). And then the

lyophilized extract was used to make the stock concentration of DG at 36 mg lyophilized extract/mL. The stock solution was diluted to the desired concentration prior to the treatment. For the LC/MS analysis, 20 mg lyophilized extract of DG was dissolved in 1 mL methanol and filtered with a 0.45 μ m filter. For analysis, 7 μ L of the sample solution was injected.

2.3. HPLC (high performance liquid chromatography) analysis of DG

The HPLC analysis of DG was described in our previous study (Bae et al., 2015). Briefly, the chromatographic analysis was performed in an HPLC-integrated system (Agilent 1200 series). LC separation used a Phenomenex Luna column (Phenomenex Inc., Torrance, CA, USA) and the mobile phase is aqueous acetonitrile containing 0.05% formic acid. The ionization mode was API-ES with positive polarity. The ionization mass range was 200–1500.

2.4. Cell culture and treatment

The human neuroblastoma cell line (SH-SY5Y) was purchased from American Type Culture Collection (Manassas, VA, USA). SH-SY5Y cells were maintained in DMEM supplemented with 10% heat-inactivated FBS and 1% penicillin/streptomycin at 37 °C in a humidified atmosphere of 5% CO₂ and 95% air. 6-OHDA was dissolved in DMEM medium at 10 mM stock solution right before use and diluted to the desired concentration in treatment medium. SH-SY5Y cells were seeded at a density of 1.0×10^6 cells/mL and incubated with DG (50–200 μ g/mL) for 16 h prior to 6-OHDA added to the medium. Cells were treated with 6-OHDA at 50 μ M and then cultured for the last 8 h of DG treatment.

2.5. Experimental animals

Male C57BL/6 mice (25–30 g, eight weeks old) were purchased from the DBL Co., Ltd., (Eumsung, Chungbuk, Republic of Korea) and kept in the University Animal Care Unit for one week prior to the experiments. Mice were housed five per cage, provided food and water ad libitum and kept in a 12 h light/dark cycle (the light was on from 07:30 to 19:30 h) at constant temperature (23 ± 1 °C) and humidity ($60 \pm 10\%$). Animal treatment and maintenance were carried out in accordance with the Animal Care and Use Guidelines issued by Kyung Hee University (KHP-2012-3-13), Korea.

2.6. Behavioral tests

2.6.1. Animals group and treatment

Animals were assigned to four groups: (1) Group 1 (vehicle-treated group), (2) Group 2 (MPTP-only treated group), (3) Group 3 (MPTP+DG 50 mg/kg/day-treated group), and (4) Group 4 (MPTP+DG 100 mg/kg/day-treated group). Each group contained eight mice. DG dissolved in saline was administered orally at 50 or 100 mg/kg/day for six days. MPTP (MPTP base form; 20 mg/kg) dissolved in saline was injected intraperitoneally (i.p.) four times at 2-h interval on the last day of DG treatment.

2.6.2. Pole test

The Pole test was a useful method to measure bradykinesia in mice PD model. We performed the Pole test seven days after the last MPTP injection. The mice were held on the top of the pole (diameter 8 mm, height 55 cm, with a rough surface). The time that mice needed to turn down completely was recorded as the time to turn (T-turn). The time needed for the mice to climb down and place four feet on the floor was recorded as the time for locomotion activity (T-LA). Each trial had a cut-off limit of 30 s.

2.6.3. Rota-rod test

We performed the Rota-rod test seven days after the last MPTP injection. To assess sensorimotor coordination, the mice were evaluated in the Rota-rod task. The Rota-rod unit consists of a rotating spindle (diameter 7.3 cm) and five individual compartments to test five mice at a time. After twice daily training for two successive days (rotation speed 4 rpm on the first day and 20 rpm on the second day), the rotation speed of the test was increased to 25 rpm on the third day in a test session. The time each mouse remained on the rotating bar was recorded for three trials, at a 5 min interval and a maximum trial length of 300 s per trial. Data were presented as the mean time on the rotating bar over the three test trials.

2.7. Preparation of brain samples

After the behavioral tests, the mice were anesthetized with 50 mg/kg Zoletil (Virbac Co., TX, USA) and were rapidly perfused transcardially with PBS, followed by 4% paraformaldehyde (PFA) in 0.1 M phosphate buffer. Then, the brains were rapidly removed, post-fixed in 4% PFA, and processed for cryoprotection in 30% sucrose at 4 °C. Frozen brains were cut into 30 μ m coronal sections using a cryostat microtome (CM3000, Leica, Germany). Then, the tissues were stored in a storing solution containing glycerin, ethylene glycol, and PB at 4 °C for immunohistochemistry.

2.8. Immunohistochemistry

Fixed mesencephalic cells on cover slips and free-floating brain sections were rinsed in PBS at room temperature before immunostaining. They were pre-treated with 1% H₂O₂ in PBS for 15 min. For dopaminergic neuron detection, they were incubated with the anti-TH antibody (1:2000 dilutions) overnight at 4 °C. They were then incubated with a biotinylated anti-mouse and an anti-rabbit IgG, respectively for 1 h followed by incubation in ABC solution for 1 h at room temperature. The activity was visualized with DAB for 3 min. After every incubation step, the cells and tissues were washed three times with PBS. Finally, the mesencephalic cells on cover slips were mounted on gelatin-coated glass slides and air-dried. The free-floating brain tissues were mounted on gelatin-coated slides, dehydrated, cleared with xylene, and cover slipped using histomount medium. All of them were photographed with an optical microscope (BX51T-32F01; Olympus Optical Co., Tokyo, Japan). For quantification of the effect of DG and its fractions in the mesencephalic dopaminergic cells, TH-positive cells were counted on at least four cover slips from independent experiments for each condition. Quantification of the effect of DG in brain tissues was performed by counting the TH-positive cell numbers in the SNpc at $\times 100$ magnification under a microscope and by measuring the optical density of TH-positive fibers in the ST at $\times 40$ magnification using Image software (Bethesda, MD, USA). Data were expressed as a percentage of the value in the vehicle-treated control group.

2.9. Western blot analysis

The SH-SY5Y cell lysates were incubated in RIPA buffer (Beverly, MA, USA) supplemented with a protease inhibitor (Roche, Mannheim, Germany) according to the manufacturer's instructions. Equal amounts of the lysate protein were loaded and separated on a 15% SDS-PAGE gel. The proteins were electrophoretically transferred to a PVDF membrane and the membrane was blocked in 5% skim milk. Then the membrane was incubated in the presence of primary antibodies against the following proteins: Caspase-3, PARP and GAPDH overnight. Then the membrane was washed and incubated with the corresponding HRP-conjugated secondary antibody.

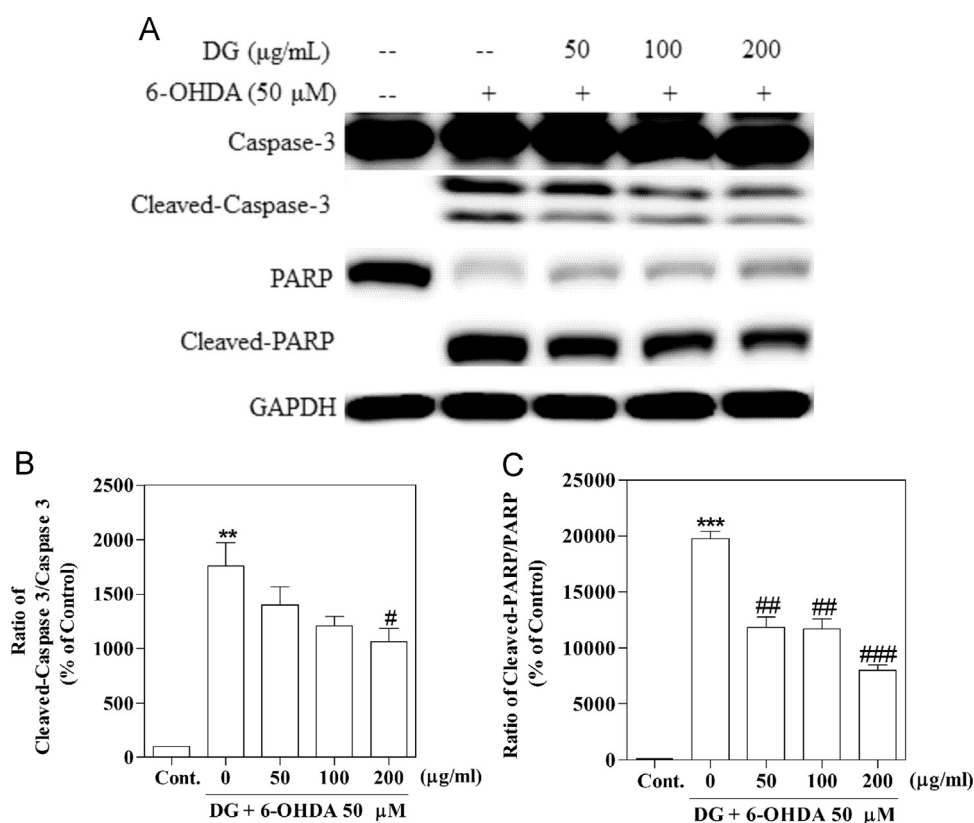


Fig. 1. Protective effects of DG on apoptosis-related protein induced by 6-OHDA in SH-SY5Y cells. Caspase-3 and PARP proteins were expressed using western blotting (A). The graphs display densitometry analyses of the expression ratios of cleaved-caspase 3/caspase-3 (B) and cleaved-PARP/PARP (C). Values are indicated as the mean $\pm$ S.E.M. ** $p < 0.01$ and *** $p < 0.001$; mean values were significantly different from the control group. # $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$; mean values were significantly different from the 6-OHDA-only-treated group.

Immunoreactive-bands were detected using an ECL detection kit and an LAS-4000 mini system (Fujifilm, Japan) was used for visualization. The intensities of the bands were normalized to the GAPDH intensity using Multi Gauge software (Fujifilm).

The SNpc tissues were lysed using a protein assay or a mitochondria/cytosol fraction kit, according to the manufacturer's instructions (Biovision Inc., Milpitas, CA, USA). Briefly, the SNpc tissues were homogenized and centrifuged at 3000 rpm for 10 min at 4 °C, the supernatant was collected, and the pellet was discarded. The supernatant was then centrifuged at 13,000 rpm for 30 min at 4 °C, after centrifugation, the supernatant was collected as the cytosolic fraction (Cyto fr.) and the pellet was resuspended, vortexed for 10 s and saved as the mitochondrial fraction (Mito fr.). The protein concentration was detected by the Bradford method. The lysates of the cytosolic fraction and the mitochondrial fraction were separated by 15% SDS-polyacrylamide gel electrophoresis, and then transferred to PVDF membranes. The membranes were incubated with 5% skim milk dissolved in TBST (25 mM Tris-Cl, 150 mM NaCl, 0.005% Tween-20) for 1 h and then with the primary antibodies (Bax, Caspase-3 and β -actin) overnight at 4 °C, and this was followed by incubation with HRP (horseradish peroxidase)-conjugated secondary antibodies for 1 h. Immunoreactive-bands were detected using an ECL detection kit and an LAS-4000 mini system (Fujifilm, Japan) was used for visualization. The intensities of the bands were normalized to the β -actin intensity using Multi Gauge software (Fujifilm).

2.10. Determination of cleaved caspase-3 activity

Cleaved caspase-3 activity was determined using commercial assay kits (Biovision, Inc.). Briefly, at the end of the experiments, the brain samples were removed and incubated by ice-cold lysis

tissue protein extraction buffer (Thermo Scientific, Rockford, IL, USA). The brain samples were then washed with ice-cold PBS and centrifuged at 3000 rpm for 5 min. The brain samples were lysed by strong vortex mixing, and the suspension was used for assays according to the manufacturer's protocols. Cleaved caspase-3 activity was normalized further to the total protein content of the cell extracts, as measured by BCA protein assay. The supernatants were collected and incubated with 4 mM DEVD-pNA as the substrate at 37 °C. The change in absorbance at 405 nm was measured using a plate reader.

2.11. Mesencephalic primary cultured cells

Mesencephalic cell cultures were derived from 14-day embryos of Sprague–Dawley rat (DBL Co., Ltd.). Mesencephalons were dissected, collected, dissociated, and seeded in Poly-L-lysine pre-coated 24-well plates at a density of 1.0×10^5 cells/well. Cultures were maintained in a humidified incubator of 5% CO₂ at 37 °C in minimal essential medium with 6 g/L glucose, 2 mM glutamine, and 10% FBS. On day *in vitro* 6, cells were treated with DG and its fractions for 1 h and stressed with 15 μ M MPP⁺ overnight. An equal volume of vehicles was given to the control and toxin groups. Then, cells were fixed with 4% PFA at room temperature for 30 min. The cells were stored in PBS at 4 °C for immunohistochemistry.

2.12. Statistical analysis

Data were analyzed with Prism 5.0 software (GraphPad Software, Inc., San Diego, CA, USA) and are expressed as the means $\pm$ S.E.M. Statistical analyses were performed using one-way analysis of variance (ANOVA) followed by the Newman–Keuls test. Statistical significance was set at $p < 0.05$.

3. Results

3.1. LC–MS identification

LC/MS profiling data of the DG extract were obtained by comparison of ESIMS data about compounds which were isolated from *Polygala tenuifolia* (Chen et al., 2001; Miyase et al., 1991, 1992; Miyase and Ueno, 1993). Relative content (%) was computed by peak areas, which were recorded at a 320 nm wavelength. The chemicals utilized were 3',6'-Disinapoyl sucrose (10.34%), Tenuifolioside A (3.92%), C (3.11%) and Tenuifolioside A (4.52%), B (4.16%), D (2.82%), H (2.49%), J (2.81%), which were reported by our previous study (Bae et al., 2015).

3.2. Protective effects of DG on apoptosis-related protein induced by 6-OHDA in SH-SY5Y

To determine the effects of DG on 6-OHDA-induced apoptosis, we measured caspase-3 and PARP expression levels in SH-SY5Y cells using western blotting (Fig. 1A). 6-OHDA-induced toxicity significantly cleaved-caspase 3 and cleaved-PARP protein levels, while DG treatment can protect SH-SY5Y cells. The expression level of cleaved-caspase 3 was significantly increased in the 6-OHDA group by $1759.16 \pm 210.48\%$ compared with the control group (Fig. 1B, $p < 0.01$). The 6-OHDA+DG 50, 100 and 200 $\mu\text{g/mL}$ groups treatment can decrease the cleaved-caspase 3 activity and the DG200 $\mu\text{g/mL}$ group decreased by $1059.79 \pm 126.89\%$ compared with the 6-OHDA group ($p < 0.05$). The expression level of cleaved-PARP was significantly increased in the 6-OHDA group by $19747.60 \pm 657.40\%$ compared with the control group (Fig. 1C, $p < 0.001$). The 6-OHDA+DG 50, 100 and 200 $\mu\text{g/mL}$ groups

treatment can significantly decrease the cleaved-PARP activity by $11693.13 \pm 941.21\%$, $11841.50 \pm 860.05\%$ and $8024.23 \pm 472.71\%$, respectively, compared with the 6-OHDA group ($p < 0.01$, $p < 0.01$, and $p < 0.001$).

3.3. Protective effects of DG against MPTP-induced behavioral impairment in mice

To investigate the effect of DG on MPTP-induced bradykinesia, a Pole test and a Rota-rod test were performed. T-turn and T-LA were significantly prolonged to $260.21 \pm 41.43\%$ and $151.82 \pm 8.57\%$ on the seventh day compared with the control group, respectively (Fig. 2A and B, $p < 0.01$). However, T-turn and T-LA were significantly shortened in the MPTP+DG 50 mg/kg group to $163.10 \pm 10.13\%$ (T-turn time) and $110.51 \pm 3.78\%$ (T-LA time) and in the MPTP+DG 100 mg/kg group, the T-turn and T-LA time were shortened to $143.05 \pm 10.67\%$ and $100.87 \pm 3.17\%$ on the seventh day compared with the MPTP group, respectively ($p < 0.05$, $p < 0.01$, and $p < 0.001$). In the Rota-rod test, we found MPTP administration significantly decreased the retention time by $18.06 \pm 5.36\%$ on the seventh day compared with control group (Fig. 2C, $p < 0.001$). However, retention times were significantly increased in the MPTP+DG 100 mg/kg group to $76.17 \pm 19.58\%$ on the seventh day compared with the MPTP group ($p < 0.05$).

3.4. Protective effects of DG against MPTP-induced dopaminergic neuron cell loss in the ST and SNpc of brain in mice

To confirm the protective effects of DG on dopaminergic neuronal damage induced by MPTP, we performed TH-IHC in the ST and SNpc (Fig. 3A). As shown in Fig. 3B and C, optical density and

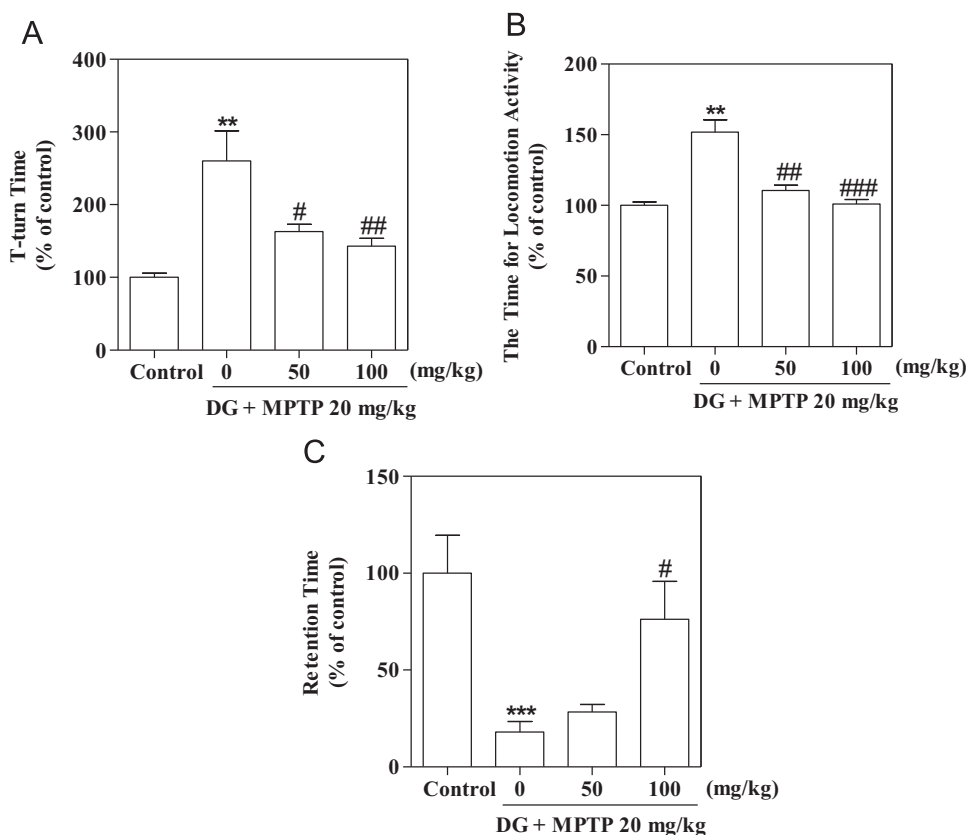


Fig. 2. Protective effects of DG against MPTP-induced behavioral impairment in mice. Saline or DG at 100 mg/kg/day were administered orally for six days, and MPTP 20 mg/kg was injected i.p. four times at 2 h intervals on the last day of DG treatment. The time taken for the mice to turn completely downward (A; T-turn) and the time taken to reach the floor (B; T-LA) were recorded. Additionally, the length of time on the rota-rod was recorded (C). Values are indicated as the mean $\pm$ SEM. ** $p < 0.01$ and *** $p < 0.001$ compared with the control group, ### $p < 0.001$, ## $p < 0.01$ and #### $p < 0.001$ compared with the MPTP-only treated-group.

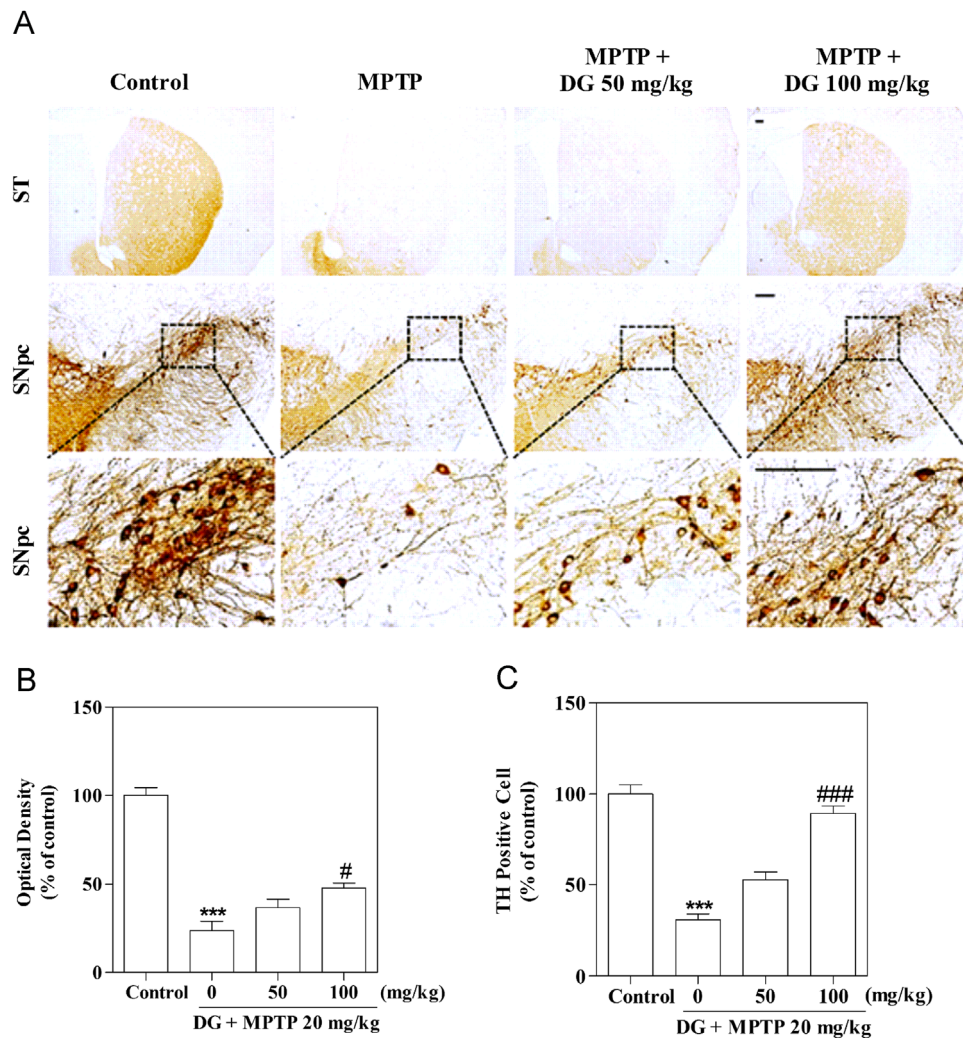


Fig. 3. Protective effects of DG against MPTP-induced dopaminergic neuron cells loss in the ST and SNpc of brain in mice. Saline or DG at 100 mg/kg/day were administered orally for seven days, and MPTP 20 mg/kg was injected i.p. four times at 2 h intervals on the last day of DG treatment. After the behavioral impairment, dopaminergic neurons were determined using TH-immunohistochemistry. Representative images are shown of the SNc and the ST of each group (A). The optical density of TH-positive fibers in the striatum was measured (B) and the numbers of TH-positive neurons in the SNc were counted (C). Values are indicated as the mean $\pm$ SEM. *** $p < 0.001$ compared with the control group, # $p < 0.05$ and ### $p < 0.001$ compared with the MPTP-only-treated group. Scale bar = 100 μ m.

TH-positive cells were observed in the control groups, whereas reduced dopaminergic neurons were found in the MPTP group, respectively, by $23.80 \pm 5.09\%$ and $30.82 \pm 3.22\%$ compared with the control group ($p < 0.001$). However, in the DG treatment 100 mg/kg group, the loss of dopaminergic neuron cells decreased significantly by $47.93 \pm 2.74\%$ and $89.19 \pm 4.10\%$ both in optical density and TH-positive cell count, compared with the MPTP group, respectively ($p < 0.05$ and $p < 0.001$).

3.5. Protective effects of DG on apoptosis-related protein induced by MPTP in mice

To determine the effects of DG on MPTP-induced apoptosis, we measured Bax and cleaved caspase-3 expression levels in the SNpc of mouse brain using western blotting (Fig. 4A). MPTP-induced toxicity significantly increased Bax and cleaved caspase-3 protein levels, while DG protected dopaminergic neurons. The expression level of cleaved caspase-3 was significantly increased in the MPTP group by $207.80 \pm 3.87\%$ compared with the control group (Fig. 4B, $p < 0.001$). However, in the MPTP+DG 50 and 100 mg/kg groups treatment significantly decreased the cleaved caspase-3 activities by $165.19 \pm 4.57\%$ and $110.01 \pm 20.46\%$, respectively, compared with the MPTP group ($p < 0.05$ and $p < 0.001$).

3.6. Protective effects of DG-fractions on dopaminergic neuronal loss induced by MPP⁺ in primary dopaminergic neurons

To examine the protective effects of DG-fractions against MPP⁺ toxicity in primary dopaminergic neurons (Fig. 5A), we counted TH-IR cell bodies. MPP⁺ neurotoxicity was defined as a $41.05 \pm 2.39\%$ reduction in the survival rate (Fig. 5B, $p < 0.001$) compared with the control group, whereas treatment of DG-fractions *T* at 0.1 and 1 μ g/mL protected them by $53.45 \pm 0.61\%$ and $55.77 \pm 7.44\%$ increase of the survival rates compared with the MPP⁺ group, respectively ($p < 0.01$ and $p < 0.001$).

4. Discussion

From the results, it was demonstrated that DG could attenuate behavioral abnormalities that were induced by MPTP in PD mimic mouse model. Additionally, the DG had a preventive effect related to the protection of nigrostriatal dopaminergic neurons through the anti-apoptosis effect from MPTP-induced neurotoxicity in the brain and as well protect the SH-SY5Y cells from the neural damage of 6-OHDA treatment.

In the present study, pathogenesis of PD involves strong

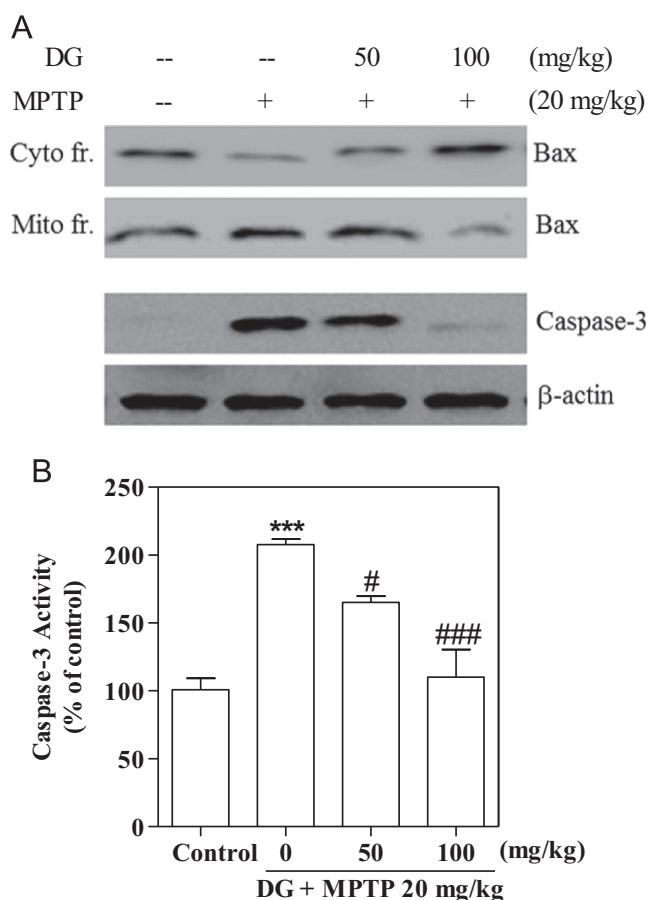


Fig. 4. Protective effects of DG on apoptosis-related protein induced by MPTP in mice. Bax in the cytosolic and mitochondrial fraction of SNpc tissues and caspase-3 proteins of mice brain were expressed using western blotting (A). The graphs display densitometry analyses of the expression ratios of caspase-3 (B). Values are indicated as the mean $\pm$ S.E.M. Cyto fr.: cytosolic fraction, Mito fr.: mitochondrial fraction. *** $p < 0.001$; mean values were significantly different from the control group. # $p < 0.05$ and ### $p < 0.001$; mean values were significantly different from the MPTP-only-treated group.

oxidative stress, reduced antioxidant levels and mitochondrial defects all known to induce apoptosis in several cellular systems. 6-OHDA was one of the widely used neurotoxins in experimentally model nigral degeneration *in vitro* as well as *in vivo* (Blum et al., 2001). Several studies have confirmed the ability of 6-OHDA to produce oxidative stress *in vitro* (Lotharius et al., 1999). Mitogen activated protein kinases (MAPKs), especially p38 MAPK, are activated in response to oxidative stress and responsible for neuronal cell death (Kim and Choi, 2010). In our study, the neural toxic effect of 6-OHDA was accompanied by the activation of caspase-3 and PARP, and the DG treatment attenuated this damage.

The treatment of MPTP produced behavioral abnormalities and, the motor incoordination, evaluated by the behavioral tests, the Rota-rod test, and the Pole test in mice. The behavioral deficits induced by MPTP were significantly suppressed by administration of DG, in particular, DG pre-treatment significantly increased motor coordination relative to the MPTP-only-treated mice. MPTP-intoxicated mice showed higher or lower values than the animals in the control group. The Pole test and the Rota-rod test are well-known methods to assess motor coordination, balance, and motor learning functions in rodents. Different dosages of DG treatment shortened the T-turn and T-LA time and increased the retention time on the rod, exhibiting the protective effect in the Pole test and Rota-rod test. The mice receiving the treatment of DG exhibited significantly less impaired mobile abilities than the MPTP-intoxicated mice.

SNpc is a special part in the midbrain that is abundant in dopaminergic neurons. An important neurotransmitter, dopamine, was secreted. Dopamine is responsible for transmitting signals between the SNpc and multiple brain regions. The connection between the SNpc and the striatum is critical for producing smooth, purposeful movement. Loss of dopamine in this circuit results in abnormal nerve-firing patterns within the brain that cause impaired movements. In the MPTP mouse model, the dopaminergic neurons in the SNpc were severely depleted, caused by MPP⁺ which was converted from MPTP and the mice with MPTP injected exhibited a serious movement abnormality. From the above, through the behavioral tests, it can be suggested that DG has an improvement effect to adjust the abnormal movements in MPTP-treated mice. The DG must have a protective effect on the SNpc dopaminergic neurons that control the movement. The SNpc was dissected for further tests to illustrate the possible mechanism of the DG protective effect.

Tyrosine hydroxylase (TH) is the rate-limiting step in the biosynthesis of DA and TH proteins, and responsible for conversion of L-dopa to dopamine (Daubner et al., 2011). The measurement of TH-immunoreactivity is thus a measure of the functionality of dopaminergic neurons present in the SNpc (Fukuda et al., 1999). TH expression is decreased in the nigrostriatal DA neurons of PD patients (Joyce et al., 1997). Comparisons of TH immunoreactivity in the ST and SNpc are performed among the four groups of animals. A marked decrease in the number of TH-positive cells is observed in the MPTP-intoxicated group. The data showed that MPTP treatment in mice induced significant damages to the DA system in SNpc, while DG treatment can inhibit the reduction in positive TH-neurons in SNpc typically observed in vehicle MPTP mice. Our study suggested that DG might protect against dopaminergic neurodegeneration induced by the neurotoxicity of MPTP. Some bioactive compounds from herbal medicine have been known to provide a significant neuroprotective effect in PD models and because DG contains many constituents with a neuroprotective effect, it is a logical next step to explore the active constituents in DG.

It is believed that the main mechanism of dopaminergic neuronal loss in MPTP-treated animals is oxidative stress generated in response to MPTP exposure (Przedborski and Vila, 2001), and the oxidative stress can induce many types of cell degradation and death. The most related form of apoptosis is induced by oxidative stress in Parkinson's disease. In the present study, we determined that DG protected dopaminergic neurons against MPTP toxicity in an *in vivo* PD model through anti-apoptosis. A key marker of apoptosis is the activity of caspase-3. With the injection of MPTP, the activity of caspase-3 was observably increased in SNpc as it induced the neuron loss and destroyed the function of SNpc. When pre-treated with DG, the activity of caspase-3 was significantly reduced and protected the dopaminergic neurons from oxidative stress. Mitochondrial dysfunction is one of the key pathogenic events of PD, and has been recognized as one of key initiators of cell apoptosis (Chakraborty et al., 2013; Jiang et al., 2013). The reduction of the membrane potential of mitochondria provides an important clue about the functional loss of mitochondria (Joshi and Bakowska, 2011) and plays a critical role in the intrinsic apoptosis pathway. Bax, a pro-apoptotic factor, belongs to the Bcl-2 family and influences the mitochondrial outer membrane permeability and apoptotic susceptibility (Kelekar and Thompson, 1998). After administration, MPTP converts to MPP⁺, MPP⁺ can pass the blood-brain barrier and promote membrane insertion of Bax and allow cytochrome c redistribution from mitochondria to cytosol, leading to further activation of the down-stream effector caspase-3 (Circu and Aw, 2010). In other words, after the treatment of DG, the function of mitochondrial was maintained and inhibited the constitution of the mitochondrial outer membrane

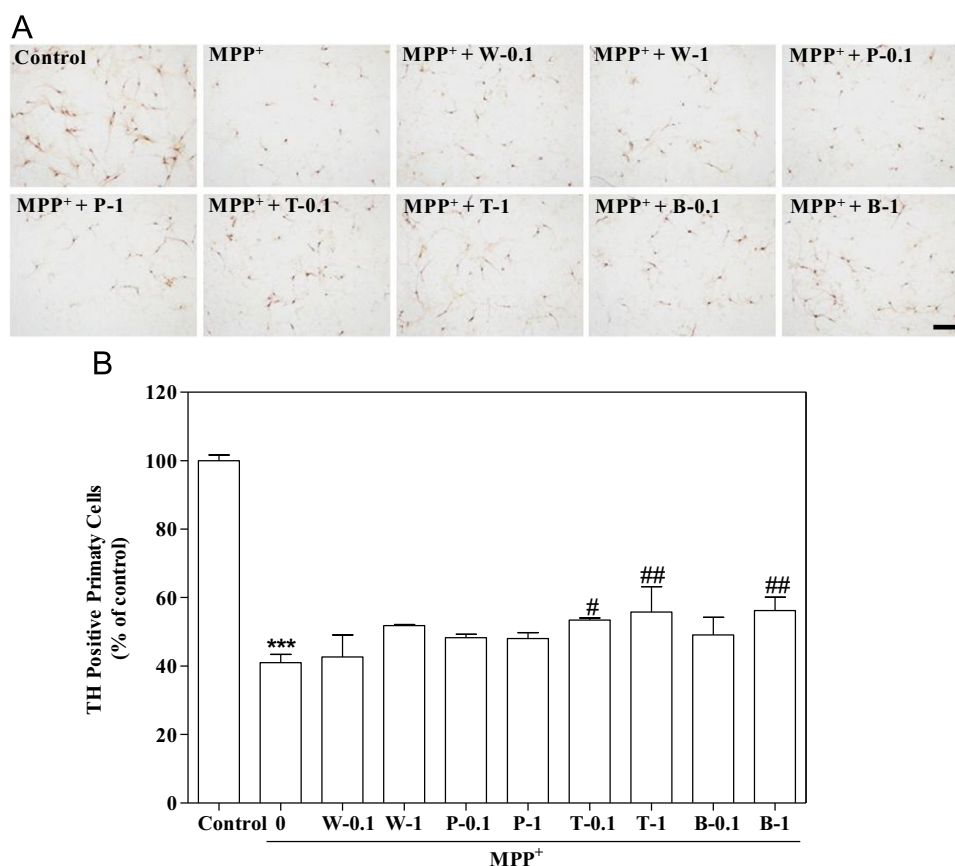


Fig. 5. Protective effects of DG-fractions on dopaminergic neuronal loss induced by MPP⁺ in primary dopaminergic neurons. Effects of DG on MPP⁺-induced toxicity in primary dopaminergic neurons. Cells were treated with DG at 0.1 and 1 μ g/mL, or vehicle for 1 h before exposure to 15 μ M MPP⁺ for 24 h. The representative images are shown (A), and the numbers of TH-positive neurons were counted (B). Values are given as the mean $\pm$ S.E.M. *** $p < 0.001$ compared to the control group, # $p < 0.05$ and ## $p < 0.01$, compared with MPP⁺-alone group. scale bar = 100 μ m.

permeabilization and reduced the damage of MPP⁺ to mitochondria. It may be a possible mechanism for DG to exert its protective effect on dopaminergic neurons. In the next step, we will detect the regulators of apoptosis and oxidative factors to illustrate the exactly mechanism of DG.

5. Conclusion

The present results indicate that DG attenuates MPTP-induced neuronal apoptotic death in a PD mouse model, at least in part, by down-regulating apoptotic signals, such as Bax expression, cytosolic cytochrome c release, and caspase-3 activation. It is suggested that the apoptosis was involved in the neuroprotective effect of DG. In the next step of this research, the anti-apoptotic mechanisms of DG will be deeply studied as a therapy strategy for PD.

Conflicts of interest

The authors declare that there are no conflicts of interest.

Acknowledgments

This work was supported by the Korea Institute of Science and Technology institutional program (2Z04381), the Bio-Synergy Research Project (NRF-2012M3A9C4048793) and the Bio & Medical Technology development Program (NRF-2015M3A9A5030735) by

the Ministry of Science, ICT and Future Planning, Korea government.

References

- Bae, N., Chung, S., Kim, H.J., Cha, J.W., Oh, H., Gu, M.Y., Oh, M.S., Yang, H.O., 2015. Neuroprotective effect of modified Chungsimyeolda-tang, a traditional Korean herbal formula, via autophagy induction in models of Parkinson's disease. *J. Ethnopharmacol.* 159, 93–101.
- Banerjee, R., Starkov, A.A., Beal, M.F., Thomas, B., 2009. Mitochondrial dysfunction in the limelight of Parkinson's disease pathogenesis. *Biochim. Biophys. Acta* 1792, 651–663.
- Barrett, P.J., Timothy Greenamyre, J., 2015. Post-translational modification of α -synuclein in Parkinson's disease. *Brain Res.* (in press).
- Bergman, H., Deuschl, G., 2002. Pathophysiology of Parkinson's disease: from clinical neurology to basic neuroscience and back. *Mov. Disord.* 17 (Suppl. 3), S28–S40.
- Blum, D., Torch, S., Lambeng, N., Nissou, M.-F., Benabid, A.-L., Sadoul, R., Verna, J.-M., 2001. Molecular pathways involved in the neurotoxicity of 6-OHDA, dopamine and MPTP: contribution to the apoptotic theory in Parkinson's disease. *Prog. Neurobiol.* 65, 135–172.
- Bove, J., Prou, D., Perier, C., Przedborski, S., 2005. Toxin-induced models of Parkinson's disease. *NeuroRx* 2, 484–494.
- Chakraborty, S., Bornhorst, J., Nguyen, T.T., Aschner, M., 2013. Oxidative stress mechanisms underlying Parkinson's disease-associated neurodegeneration in *C. elegans*. *Int. J. Mol. Sci.* 14, 23103–23128.
- Chen, C.C., Huang, Y.L., Huang, F.I., Wang, C.W., Ou, J.C., 2001. Water-soluble glycosides from *Ruta graveolens*. *J. Nat. Prod.* 64, 990–992.
- Choi, J.G., Kim, H.G., Kim, M.C., Yang, W.M., Huh, Y., Kim, S.Y., Oh, M.S., 2011. *Polygalae* radix inhibits toxin-induced neuronal death in the Parkinson's disease models. *J. Ethnopharmacol.* 134, 414–421.
- Circu, M.L., Aw, T.Y., 2010. Reactive oxygen species, cellular redox systems, and apoptosis. *Free Radic. Biol. Med.* 48, 749–762.
- Daubner, S.C., Le, T., Wang, S., 2011. Tyrosine hydroxylase and regulation of dopamine synthesis. *Arch. Biochem. Biophys.* 508, 1–12.
- Dauer, W., Przedborski, S., 2003. Parkinson's disease: mechanisms and models.

- Neuron 39, 889–909.
- Fukuda, T., Takahashi, J., Tanaka, J., 1999. Tyrosine hydroxylase-immunoreactive neurons are decreased in number in the cerebral cortex of Parkinson's disease. *Neuropathology* 19, 10–13.
- Hartmann, A., Hunot, S., Michel, P.P., Muriel, M.P., Vyas, S., Faucheux, B.A., Mouatt-Prigent, A., Turmel, H., Srinivasan, A., Ruberg, M., Evan, G.I., Agid, Y., Hirsch, E.C., 2000. Caspase-3: a vulnerability factor and final effector in apoptotic death of dopaminergic neurons in Parkinson's disease. *Proc. Natl. Acad. Sci. USA* 97, 2875–2880.
- Hartmann, A., Troade, J.D., Hunot, S., Kikly, K., Faucheux, B.A., Mouatt-Prigent, A., Ruberg, M., Agid, Y., Hirsch, E.C., 2001. Caspase-8 is an effector in apoptotic death of dopaminergic neurons in Parkinson's disease, but pathway inhibition results in neuronal necrosis. *J. Neurosci.* 21, 2247–2255.
- Jazvinskac Jembrek, M., Hof, P.R., Simic, G., 2015. Ceramides in Alzheimer's disease: key mediators of neuronal apoptosis induced by oxidative stress and A β accumulation. *Oxid. Med. Cell. Longev.* 2015, 346783.
- Jiang, J., Jiang, J., Zuo, Y., Gu, Z., 2013. Rapamycin protects the mitochondria against oxidative stress and apoptosis in a rat model of Parkinson's disease. *Int. J. Mol. Med.* 31, 825–832.
- Jiang, J., Meng, F.Y., He, Z., Ning, Y.L., Li, X.H., Song, H., Wang, J., Zhou, R., 2014. Sulfated modification of *longan* polysaccharide and its immunomodulatory and antitumor activity *in vitro*. *Int. J. Biol. Macromol.* 67, 323–329.
- Joshi, D.C., Bakowska, J.C., 2011. Determination of mitochondrial membrane potential and reactive oxygen species in live rat cortical neurons. *J. Vis. Exp.* 51, 1–4.
- Joyce, J.N., Smutzer, G., Whitty, C.J., Myers, A., Bannon, M.J., 1997. Differential modification of dopamine transporter and tyrosine hydroxylase mRNAs in midbrain of subjects with Parkinson's, Alzheimer's with parkinsonism, and Alzheimer's disease. *Mov. Disord.* 12, 885–897.
- Ka, M.H., Choi, E.H., Chun, H.S., Lee, K.G., 2005. Antioxidative activity of volatile extracts isolated from *Angelica tenuissimae* roots, peppermint leaves, pine needles, and sweet flag leaves. *J. Agric. Food Chem.* 53, 4124–4129.
- Kelekar, A., Thompson, C.B., 1998. Bcl-2-family proteins: the role of the BH3 domain in apoptosis. *Trends Cell. Biol.* 8, 324–330.
- Kim, E.K., Choi, E.J., 2010. Pathological roles of MAPK signaling pathways in human diseases. *Biochim. Biophys. Acta* 1802, 396–405.
- Leem, K.H., Park, H.K., 2007. Traditional Korean medicine: now and the future. *Neurol. Res.* 29 (Suppl. 1), S3–S4.
- Lotharius, J., Dugan, L.L., O'Malley, K.L., 1999. Distinct mechanisms underlie neurotoxin-mediated cell death in cultured dopaminergic neurons. *J. Neurosci.* 19, 1284–1293.
- Lv, J., Jia, H., Jiang, Y., Ruan, Y., Liu, Z., Yue, W., Beyreuther, K., Tu, P., Zhang, D., 2009. Tenuifolin, an extract derived from *tenuigenin*, inhibits amyloid- β secretion *in vitro*. *Acta Physiol.* 196, 419–425.
- Lynch-Day, M.A., Mao, K., Wang, K., Zhao, M., Klionsky, D.J., 2012. The role of autophagy in Parkinson's disease. *Cold Spring Harb. Perspect. Med.* 2, a009357.
- Markey, S.P., Johannessen, J.N., Chiueh, C.C., Burns, R.S., Herkenham, M.A., 1984. Intraneuronal generation of a pyridinium metabolite may cause drug-induced parkinsonism. *Nature* 311, 464–467.
- Marshall, K.A., Daniel, S.E., Cairns, N., Jenner, P., Halliwell, B., 1997. Upregulation of the anti-apoptotic protein Bcl-2 may be an early event in neurodegeneration: studies on Parkinson's and incidental Lewy body disease. *Biochem. Biophys. Res. Commun.* 240, 84–87.
- Miyase, T., Iwata, Y., Ueno, A., 1991. Tenuifolioses A-F, oligosaccharide multi-esters from the roots of *Polygala tenuifolia* WILLD. *Chem. Pharm. Bull.* 39, 3082–3084.
- Miyase, T., Iwata, Y., Ueno, A., 1992. Tenuifolioses G-P, oligosaccharide multi-esters from the roots of *Polygala tenuifolia* WILLD. *Chem. Pharm. Bull.* 40, 2741–2748.
- Miyase, T., Ueno, A., 1993. Sucrose derivatives from the roots of *Polygala tenuifolia*. *Jpn. J. Pharmacogn.* 47, 267–278.
- Mogi, M., Harada, M., Kondo, T., Mizuno, Y., Narabayashi, H., Riederer, P., Nagatsu, T., 1996. Bcl-2 protein is increased in the brain from parkinsonian patients. *Neurosci. Lett.* 215, 137–139.
- More, S.V., Kumar, H., Kang, S.M., Song, S.Y., Lee, K., Choi, D.K., 2013. Advances in neuroprotective ingredients of medicinal herbs by using cellular and animal models of Parkinson's disease. *Evid. Based Complement. Altern. Med.* 2013, 957875.
- Naito, R., Tohda, C., 2006. Characterization of anti-neurodegenerative effects of *Polygala tenuifolia* in A β (25–35)-treated cortical neurons. *Biol. Pharm. Bull.* 29, 1892–1896.
- Przedborski, S., Jackson-Lewis, V., Djaldetti, R., Liberatore, G., Vila, M., Vukosavic, S., Almer, G., 2000. The parkinsonian toxin MPTP: action and mechanism. *Restor. Neurol. Neurosci.* 16, 135–142.
- Przedborski, S., Vila, M., 2001. MPTP: a review of its mechanisms of neurotoxicity. *Clin. Neurosci. Res.* 1, 407–418.
- Romanucci, M., Della Salda, L., 2015. Oxidative stress and protein quality control systems in the aged canine brain as a model for human neurodegenerative disorders. *Oxid. Med. Cell. Longev.* 2015, 940131.
- Schapira, A.H., Jenner, P., 2011. Etiology and pathogenesis of Parkinson's disease. *Mov. Disord.* 26, 1049–1055.
- Schmidt, N., Ferger, B., 2001. Neurochemical findings in the MPTP model of Parkinson's disease. *J. Neural Transm.* 108, 1263–1282.
- Serviddio, G., Romano, A.D., Cassano, T., Bellanti, F., Altomare, E., Vendemiale, G., 2011. Principles and therapeutic relevance for targeting mitochondria in aging and neurodegenerative diseases. *Curr. Pharm. Des.* 17, 2036–2055.
- Shim, E.B., Lee, S., Kim, J.Y., Earm, Y.E., 2008. Physiome and Sasang constitutional medicine. *J. Physiol. Sci.* 58, 433–440.
- Shimohama, S., Sawada, H., Kitamura, Y., Taniguchi, T., 2003. Disease model: Parkinson's disease. *Trends Mol. Med.* 9, 360–365.
- Smeyne, R.J., Jackson-Lewis, V., 2005. The MPTP model of Parkinson's disease. *Mol. Brain Res.* 134, 57–66.
- Subramaniam, S.R., Chesselet, M.F., 2013. Mitochondrial dysfunction and oxidative stress in Parkinson's disease. *Prog. Neurobiol.* 106–107, 17–32.
- Tseng, H.C., Wu, W.T., Huang, H.S., Wu, M.C., 2014. Antimicrobial activities of various fractions of *longan* (*Dimocarpus longan* Lour. Fen Ke) seed extract. *Int. J. Food Sci. Nutr.* 65, 589–593.
- Wang, Z., Wang, D., Li, Y., Zhang, X., 2014. Protective effects of verapamil against H₂O₂-induced apoptosis in human lens epithelial cells. *Biomol. Ther.* 22, 553–557.
- Zuo, L., Motherwell, M.S., 2013. The impact of reactive oxygen species and genetic mitochondrial mutations in Parkinson's disease. *Gene* 532, 18–23.